1 Number

What students need to learn:

Ref	Content descriptor	Concepts and skills
N a	Add, subtract, multiply and divide any number	• Add, subtract, multiply and divide whole numbers, integers, fractions, decimals and numbers in index form • Add, subtract, multiply and divide negative numbers • Multiply or divide by any number between 0 and 1 • Solve a problem involving division by a decimal (up to 2 decimal places)
N b	Order rational numbers	• Order integers, decimals and fractions • Understand and use positive numbers and negative integers, both as positions and translations on a number line
N c	Use the concepts and vocabulary of factor (divisor), multiple, common factor, Highest Common Factor, Least Common Multiple, prime number and prime factor decomposition	• Identify factors, multiples and prime numbers • Find the prime factor decomposition of positive integers • Find the common factors and common multiples of two numbers • Find the Highest Common Factor (HCF) and the Least Common Multiple (LCM) of two numbers
N d	Use the terms square, positive and negative square root, cube and cube root	• Recall integer squares from 2×2 to 15×15 and the corresponding square roots • Recall the cubes of 2, 3, 4, 5 and 10
N e	Use index notation for squares, cubes and powers of 10	• Use index notation for squares and cubes • Use index notation for integer powers of 10 • Find the value of calculations using indices
N f	Use index laws for multiplication and division of integer, **fractional and negative** powers	• Use index laws to simplify and calculate the value of numerical expressions involving multiplication and division of integer, **fractional and negative** powers, and powers of a power • **Recall that n^0 = 1 and $n^{-1} = \frac{1}{n}$ for positive integers n as well as $n^{\frac{1}{2}} = \sqrt{n}$ and $n^{\frac{1}{3}} = \sqrt[3]{n}$ for any positive number n**

Ref	Content descriptor	Concepts and skills
N g	**Interpret, order and calculate with numbers written in standard index form**	• **Use standard form, expressed in conventional notation** • **Be able to write very large and very small numbers presented in a context in standard form** • **Convert between ordinary and standard form representations** • **Interpret a calculator display using standard form** • **Calculate with standard form**
N h	Understand equivalent fractions, simplifying a fraction by cancelling all common factors	• Find equivalent fractions • Write a fraction in its simplest form • Convert between mixed numbers and improper fractions
N i	Add and subtract fractions	• Add and subtract fractions
N j	Use decimal notation and recognise that each terminating decimal is a fraction	• Recall the fraction-to-decimal conversion of familiar simple fractions • Convert between fractions and decimals
N k	Recognise that recurring decimals are exact fractions, and that some exact fractions are recurring decimals	• Recognise that recurring decimals are exact fractions, and that some exact fractions are recurring decimals • Convert between recurring decimals and fractions • **Understand a recurring decimal to fraction proof**

Higher

Ref	Content descriptor	Concepts and skills
N l	Understand that 'percentage' means 'number of parts per 100' and use this to compare proportions	• Convert between fractions, decimals and percentages
N m	Use percentage, **repeated proportional change**	• Use percentages to solve problems • Use percentages in real-life situations – VAT – Simple Interest – Income tax calculations – **Compound interest** – **Depreciation** – Find prices after a percentage increase or decrease – **Percentage profit and loss** • **Calculate an original amount when given the transformed amount after a percentage change** • **Calculate repeated proportional change**
N n	**Understand and use direct and indirect proportion**	• **Calculate an unknown quantity from quantities that vary in direct or inverse proportion**
N o	Interpret fractions, decimals and percentages as operators	• Find a fraction of a quantity • Express a given number as a fraction of another number • Find a percentage of a quantity • Use decimals to find quantities • Express a given number as a percentage of another number • Understand the multiplicative nature of percentages as operators • **Represent repeated proportional change using a multiplier raised to a power** • **Use compound interest** • Use a multiplier to increase or decrease by a percentage in any scenario where percentages are used
N p	Use ratio notation, including reduction to its simplest form and its various links to fraction notation	• Use ratios • Write ratios in their simplest form

Ref	Content descriptor	Concepts and skills
N q	Understand and use number operations and the relationships between them, including inverse operations and hierarchy of operations	• Multiply and divide numbers, using the commutative, associative, and distributive laws and factorisation where possible, or place value adjustments • Use brackets and the hierarchy of operations • Use one calculation to find the answer to another • Understand 'reciprocal' as multiplicative inverse, knowing that any non-zero number multiplied by its reciprocal is 1 (and that zero has no reciprocal, because division by zero is not defined) • Find reciprocals • Use inverse operations • **Understand that the inverse operation of raising a positive number to a power n is raising the result of this operation to the power $\frac{1}{n}$** • Understand and use unit fractions as multiplicative inverses • Solve word problems • **Use reverse percentage calculations**
N r	**Use surds and π in exact calculations**	• **Use surds and π in exact calculations, without a calculator** • **Give an answer to a question involving the area of a circle as 25π** • **Give an answer to use of Pythagoras' theorem as $\sqrt{13}$** • **Write $(3 - \sqrt{3})^2$ in the form $a + b\sqrt{3}$** • **Rationalise a denominator**

Higher

Ref	Content descriptor	Concepts and skills
N s	**Calculate upper and lower bounds**	• **Calculate the upper and lower bounds of calculations, particularly when working with measurements** • **Find the upper and lower bounds of calculations involving perimeter, areas and volumes of 2-D and 3-D shapes** • **Find the upper and lower bounds in real life situations using measurements given to appropriate degrees of accuracy** • **Give the final answer to an appropriate degree of accuracy following an analysis of the upper and lower bounds of a calculation**
N t	Divide a quantity in a given ratio	• Divide a quantity in a given ratio • Solve a ratio problem in a context
N u	Approximate to specified or appropriate degrees of accuracy including a given power of ten, number of decimal places and significant figures	• Round numbers to a given power of 10 • Round to the nearest integer and to any number of significant figures • Round to a given number of decimal places • Estimate answers to calculations, including use of rounding

Ref	Content descriptor	Concepts and skills
N v	Use calculators effectively and efficiently, including **trigonometrical** functions	• Enter a range of calculations, including those involving time and money • Know how to enter complex calculations • Use an extended range of calculator functions, including $+$, $-$, $\times$, $\div$, x^2, $\sqrt{x}$, memory, x^y, $x^{1/y}$, brackets and **trigonometric functions** • Understand, and interpret, the calculator display • Understand that premature rounding can cause problems when undertaking calculations with more than one step • **Calculate the upper and lower bounds of calculations, particularly when working with measurements** • **Use standard form display and know how to enter numbers in standard form** • **Calculate using standard form** • **Use calculators for reverse percentage calculations by doing an appropriate division** • **Use calculators to explore exponential growth and decay**